Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 01: Introduction to Data Analysis

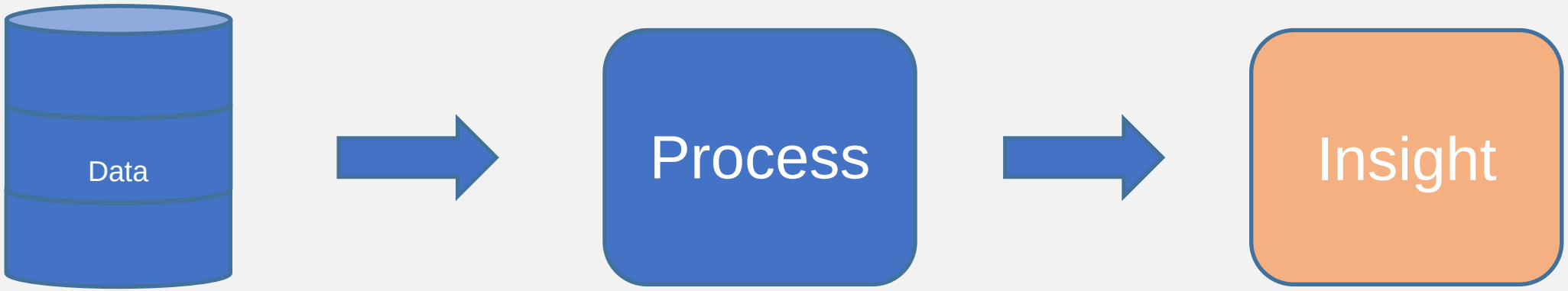
Digital for Executive

Outline

- What is Data Analysis
- Skills in Data Analysis
- Tools for Data Analysis
- Data Analysis Workflow
- Install MySQL and MySQL Workbench
- Install Jupyter Notebook

What is Data Analysis

Data Analysis is a process of inspecting, cleansing, transforming and modeling data with the goal of discovering useful information, informing conclusion and support decision making. ~Wikipedia (2020)



Structured Data

Variables (columns)

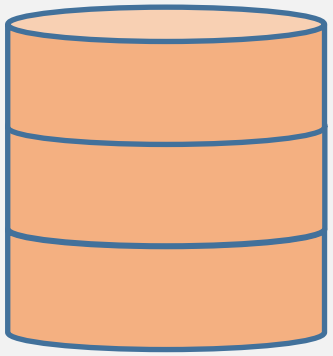
Observation (rows)

Cust ID	Age	Gender	...	Edu	Balance	...	Income	...	...	Default

Data Storage

SQL Databse

No SQL Database



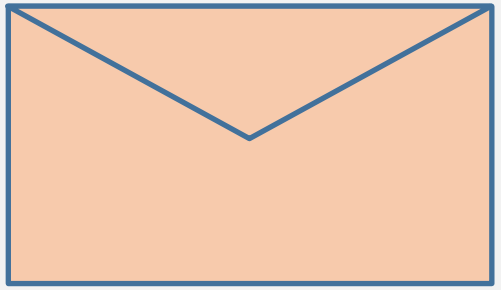
Streaming Data

JSON, XML file, etc

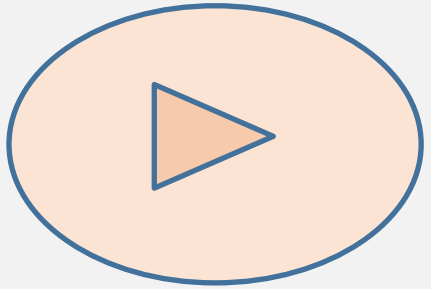
Semi-structured Data

```
<University>
  <Student ID="1">
    <Name>John</Name>
    <Age>18</Age>
    <Degree>B.Sc.</Degree>
  </Student>
  <Student ID="2">
    <Name>David</Name>
    <Age>31</Age>
    <Degree>Ph.D. </Degree>
  </Student>
....
</University>
```

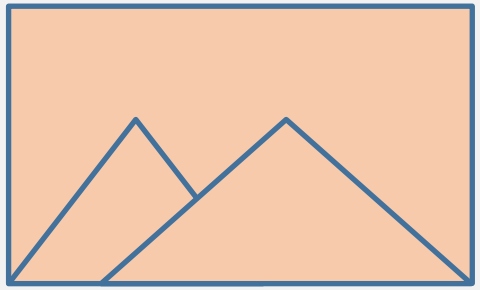
Unstructured Data



Texts

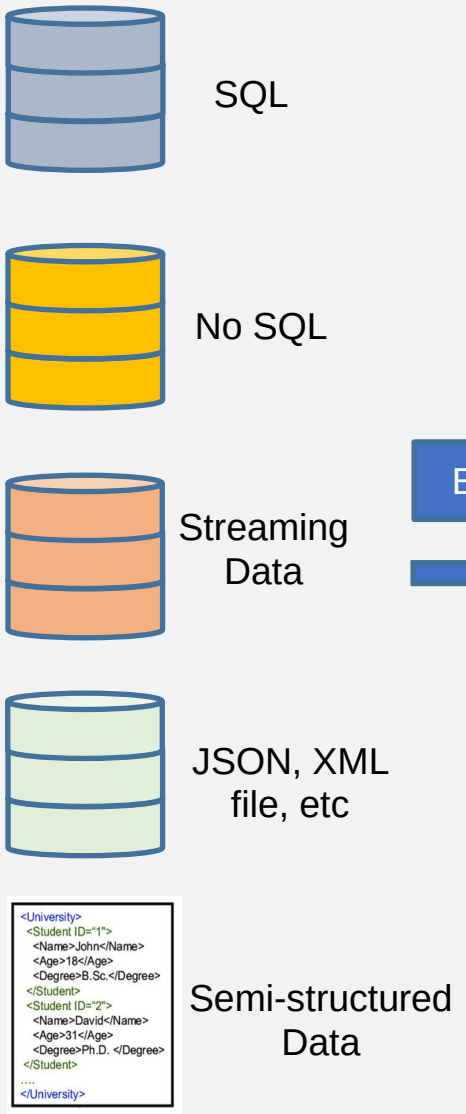


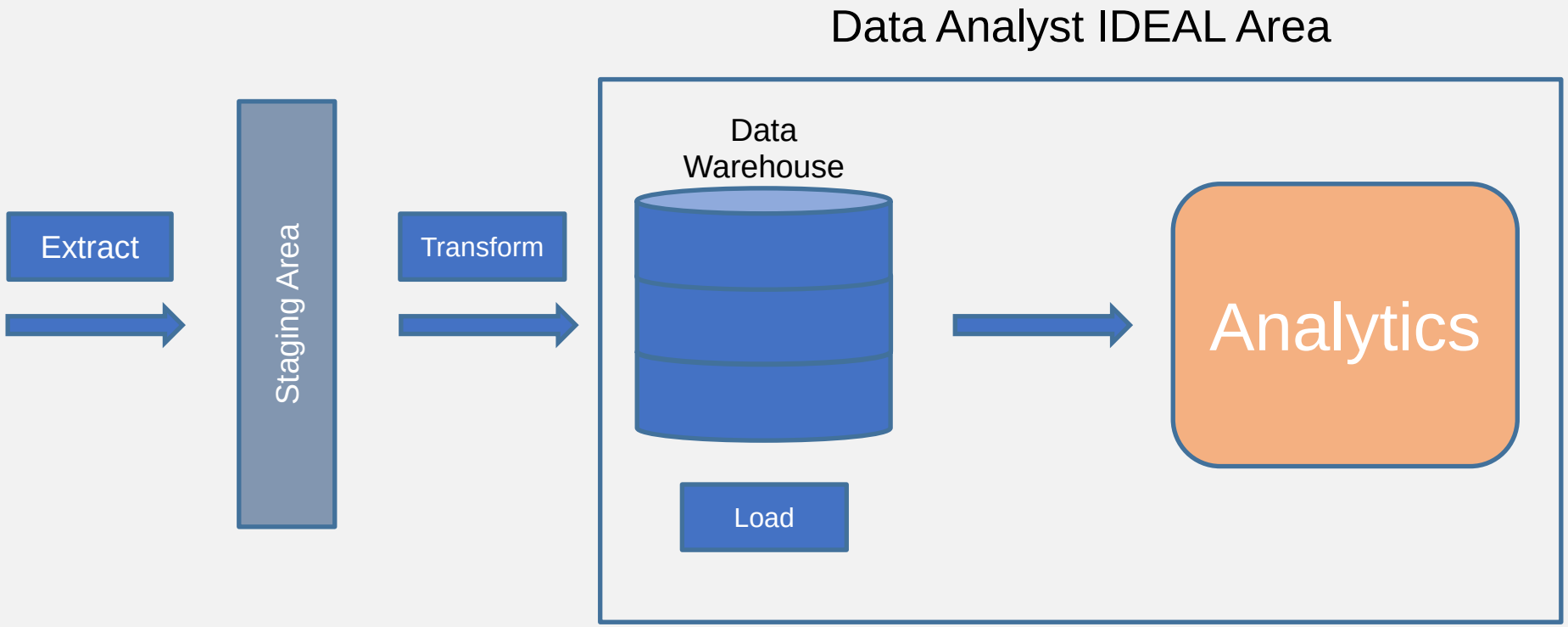
Videos



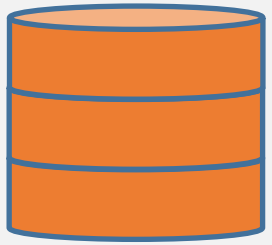
Images

Advanced Data Processing (such as ETL)

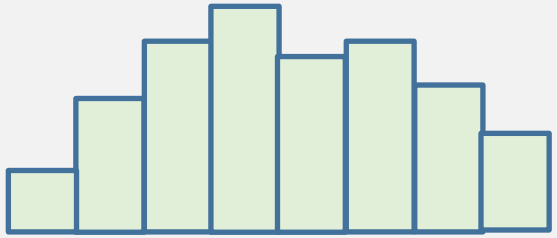




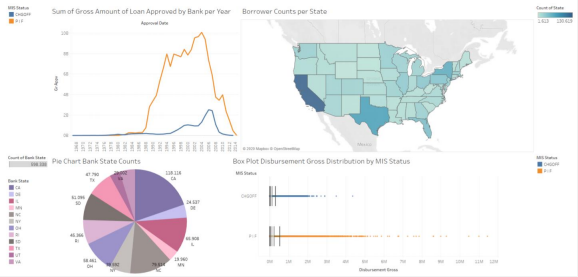
Skills in Data Analysis



SQL
(Query)



Statistics



Data
Visualization

Another Skills :

- Statistical Programming Language (Python or R)
- Data Cleaning and Preparation
- Critical Thinking
- Problem Solving
- Big Data Processing Framework

SQL (Query)

SQL Query are used for data retrieval process drom a database or a data warehous.

SQL Query

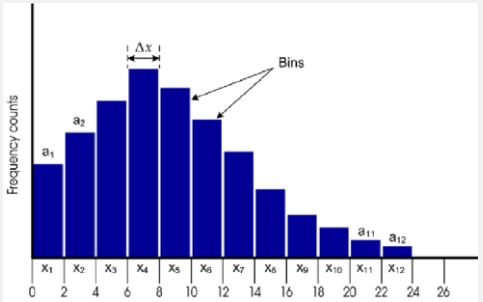
```
SELECT
    emp_no, first_name, gender, hire_date
FROM
    employees;
```

Query Result

	emp_no	first_name	gender	hire_date
▶	10001	Georgi	M	1986-06-26
	10002	Bezalel	F	1985-11-21
	10003	Parto	M	1986-08-28
	10004	Chirstian	M	1986-12-01
	10005	Kyoichi	M	1989-09-12
	10006	Anneke	F	1989-06-02
	10007	Tzvetan	F	1989-02-10
	10008	Saniya	M	1994-09-15
	10009	Sumant	F	1985-02-18
	10010	Duangkaew	F	1989-08-24

Statistics

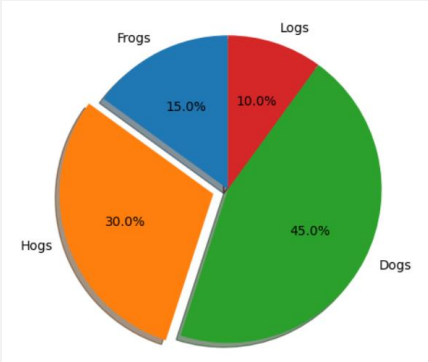
Statistics used to explore your data and drawing conclusion

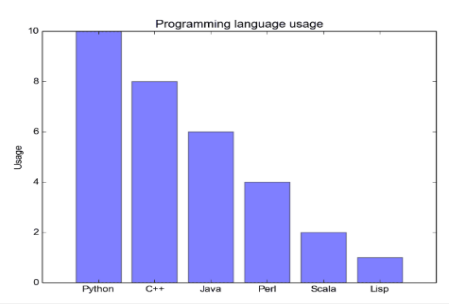



```
tips = sns.load_dataset('tips')
```

```
tips.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
25%	13.347500	2.000000	2.000000
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000



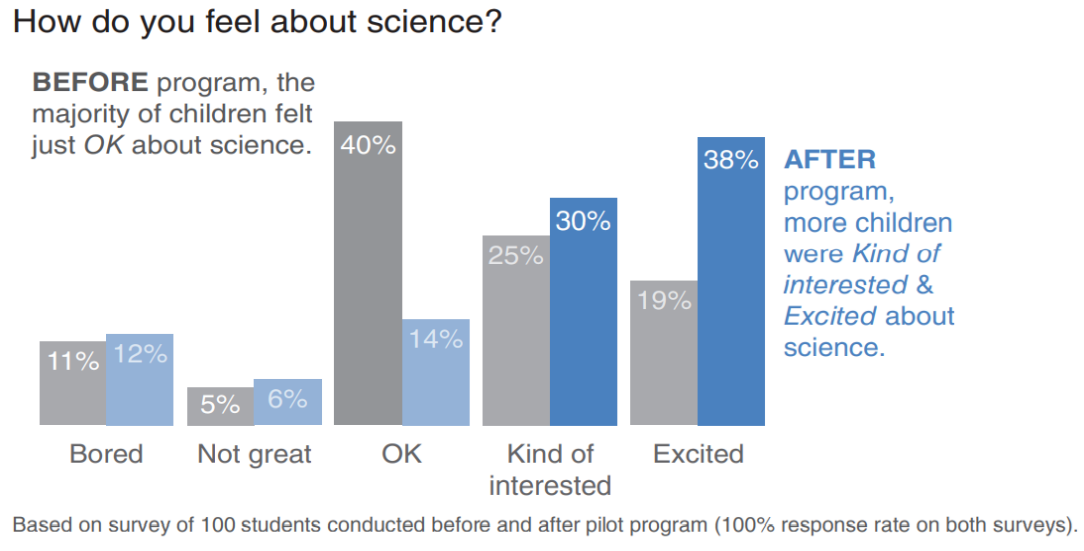


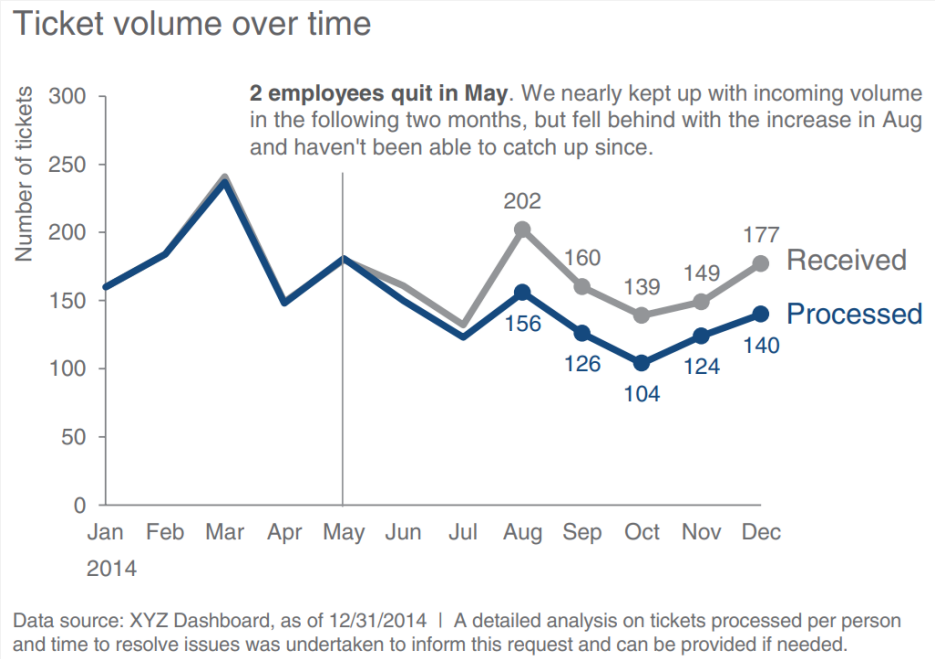
A/B Testing is hypothesis testing in disguise

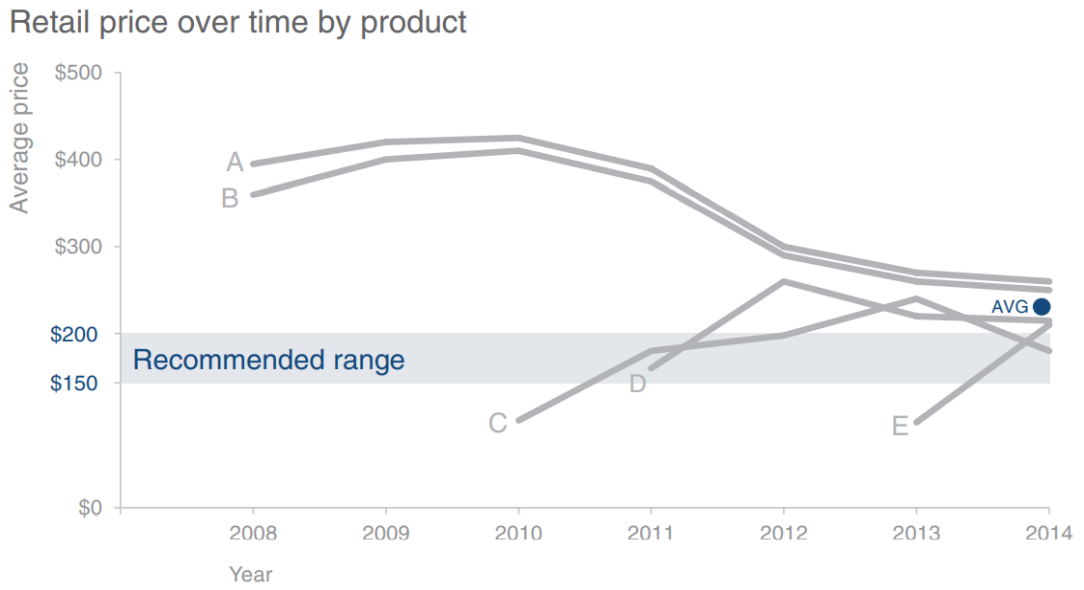
	Layout A	Layout B
Visitors	122	118
Customers	22	25
Conversion %	18.0%	21.2%

Data Visualization

share your findings with other by using data visualization







Tools For Data Analysis

Another Tools :

- Paid Statistical Analytics Software: SAS, SPSS, Minitab, STATA etc
- Ms. Excel, Spreadsheet
- Spark, Hadoop

Database

Database is a collection of information that is organized so that it can be easily accessed, managed and updated. Mostly, database use structured query language (SQL) for writing and querying the data.

Data Warehouse

A data warehouse is an enterprise system used for the analysis and reporting of structured and semi-structured data from multiple sources. ~ Google (2020).
We also use structured query language (SQL) to retrive data from a data warehouse.

TERADATA

CLOUDERA

BigQuery

amazon web services

Python

Python is a free open source tools. Python can be used for data analysis (R can be used as an alternative). It also can be integrated with any platform such as MYSQL, SQL Server, Mongo DB, BigQuery etc

```
import numpy as np

age = [22,22,23,23,29,24,27,28]

print('mean =',np.mean(age))
print('Standard Deviation =',np.std(age))
print('Q1 =',np.quantile(age,0.25))
print('Q3 =',np.quantile(age,0.75))
print('IQR =',np.quantile(age,0.75) - np.quantile(age,0.25))
```

```
mean = 24.75
Standard Deviation = 2.6339134382131846
Q1 = 22.75
Q3 = 27.25
IQR = 4.5
```

```
import seaborn as sns

tips = sns.load_dataset('tips')

tips.describe()
```

	total_bill	tip	size
count	244.000000	244.000000	244.000000
mean	19.785943	2.998279	2.569672
std	8.902412	1.383638	0.951100
min	3.070000	1.000000	1.000000
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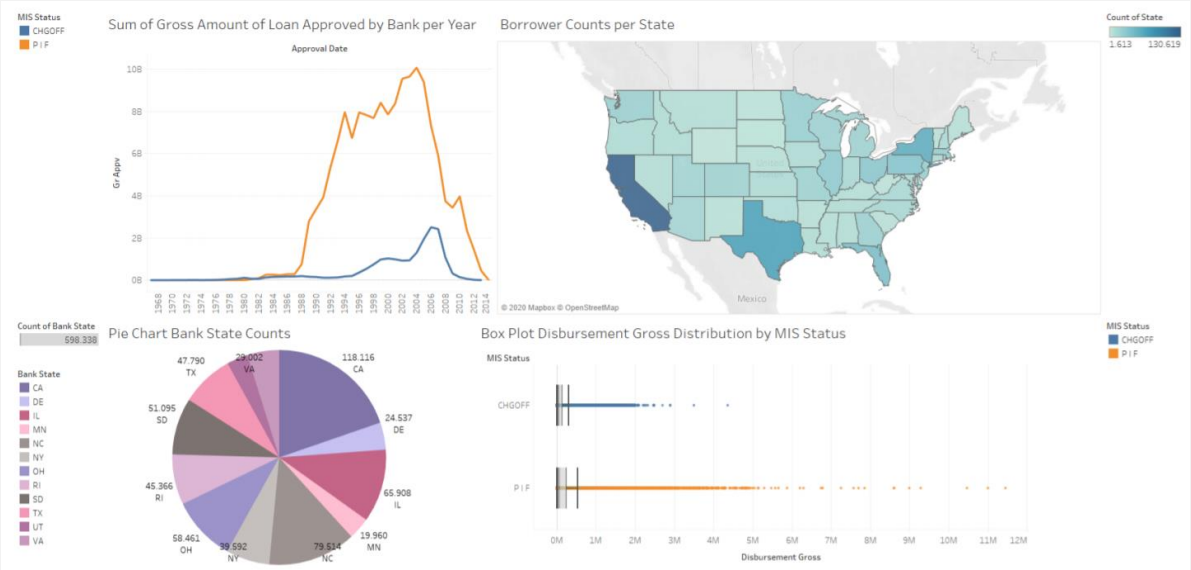
50%	17.795000	2.900000	2.000000
75%	24.127500	3.562500	3.000000
max	50.810000	10.000000	6.000000

BI Tools

BI Tools are tools that use set of methodologies and tech to prepare, present and help analyze data. You should be able to transform data into actionable insight to help decision makers.

Tableau Dashboard

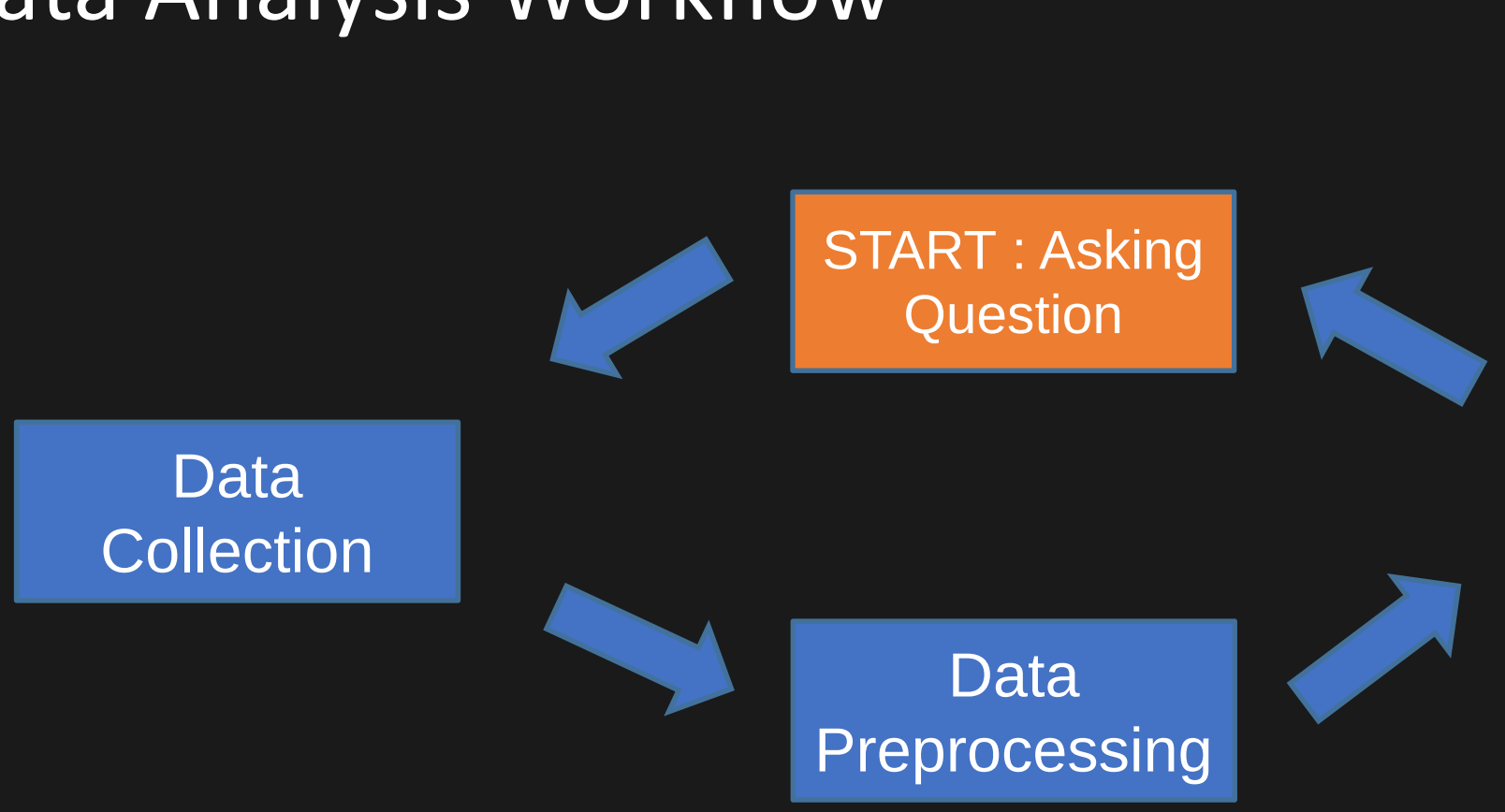


Power BI Dashboard

There are also some people start to develop some chatbot to generate any visualization

Install MySQL Workbench + MySQL Server

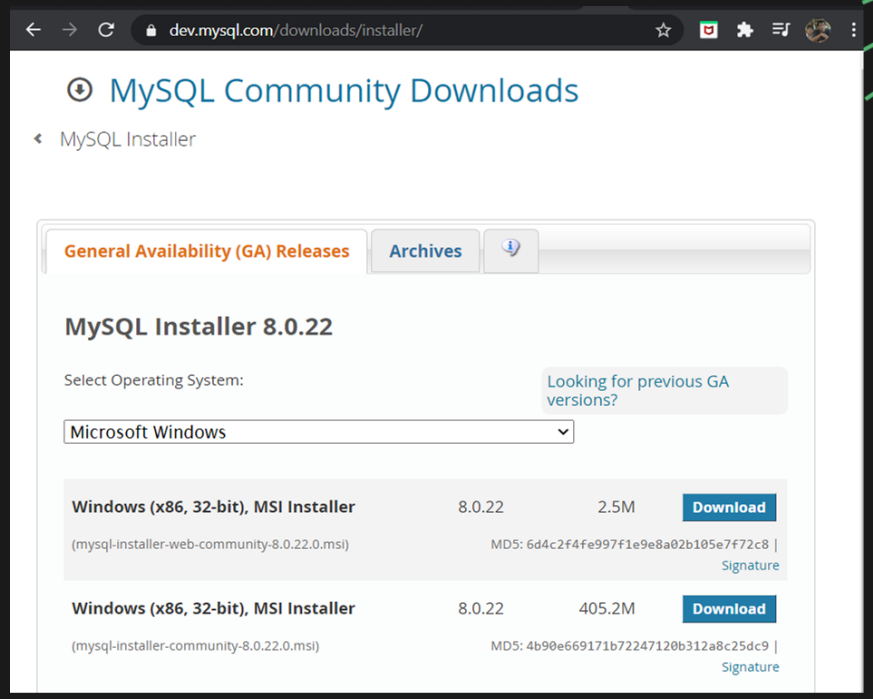
Data Analysis Workflow

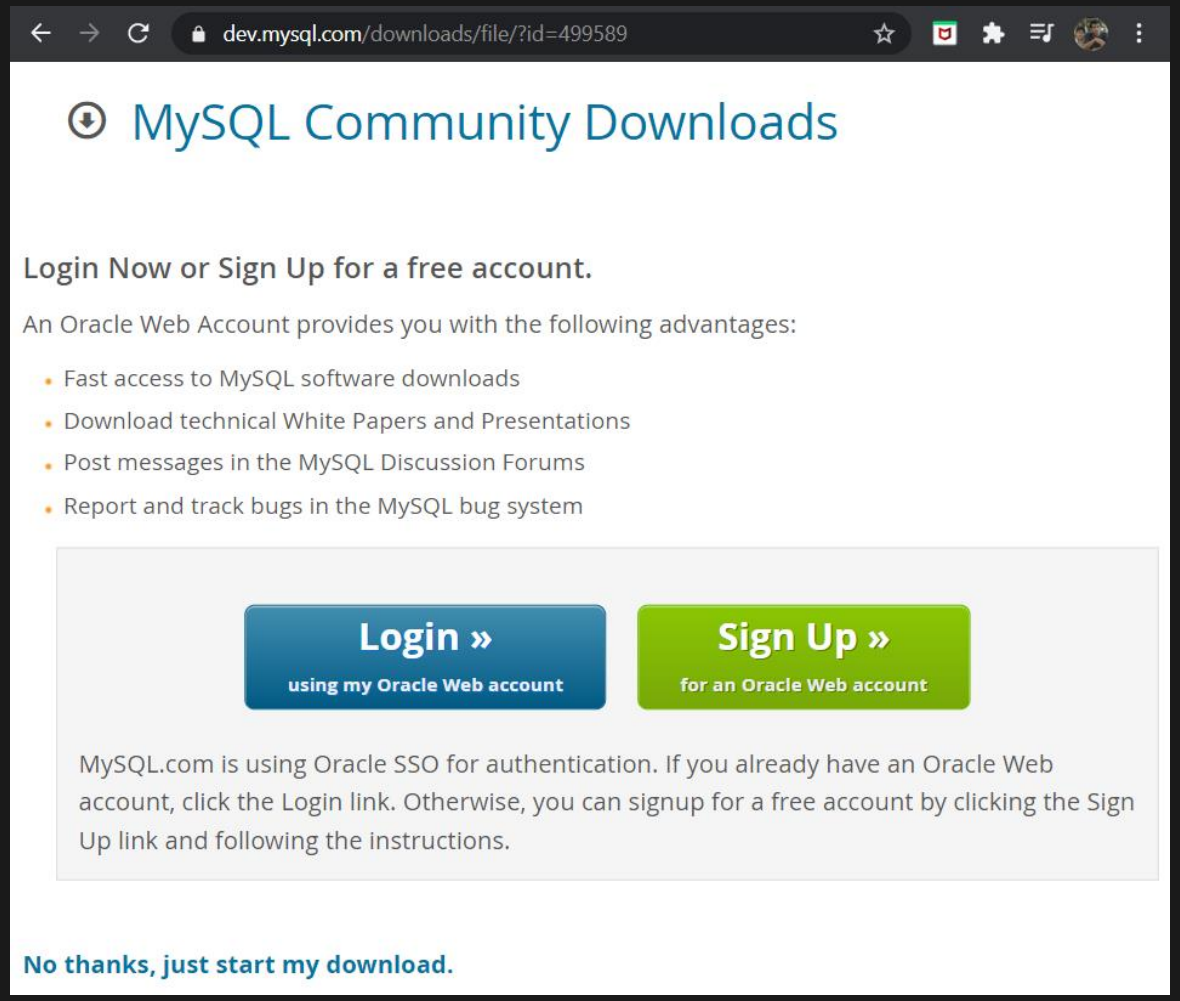


Visit MySQL website

- MySQL menyediakan 2 jenis file, yakni dengan ukuran kecil (sekitar 2.5 MB), atau yang berukuran besar (sekitar 405.2 MB).
- Versi ukuran kecil sebenarnya adalah versi online installer, yang akan mendownload seluruh file MySQL pada saat proses instalasi berlangsung dan kita harus online.
- Sedangkan yang berukuran besar adalah versi offline installer yang tidak perlu terkoneksi ke internet ketika proses instalasi.
- Agar lebih mudah kita akan gunakan online installer. Klik tombol Download untuk melanjutkan.

Visit https://dev.mysql.com/downloads/installer/

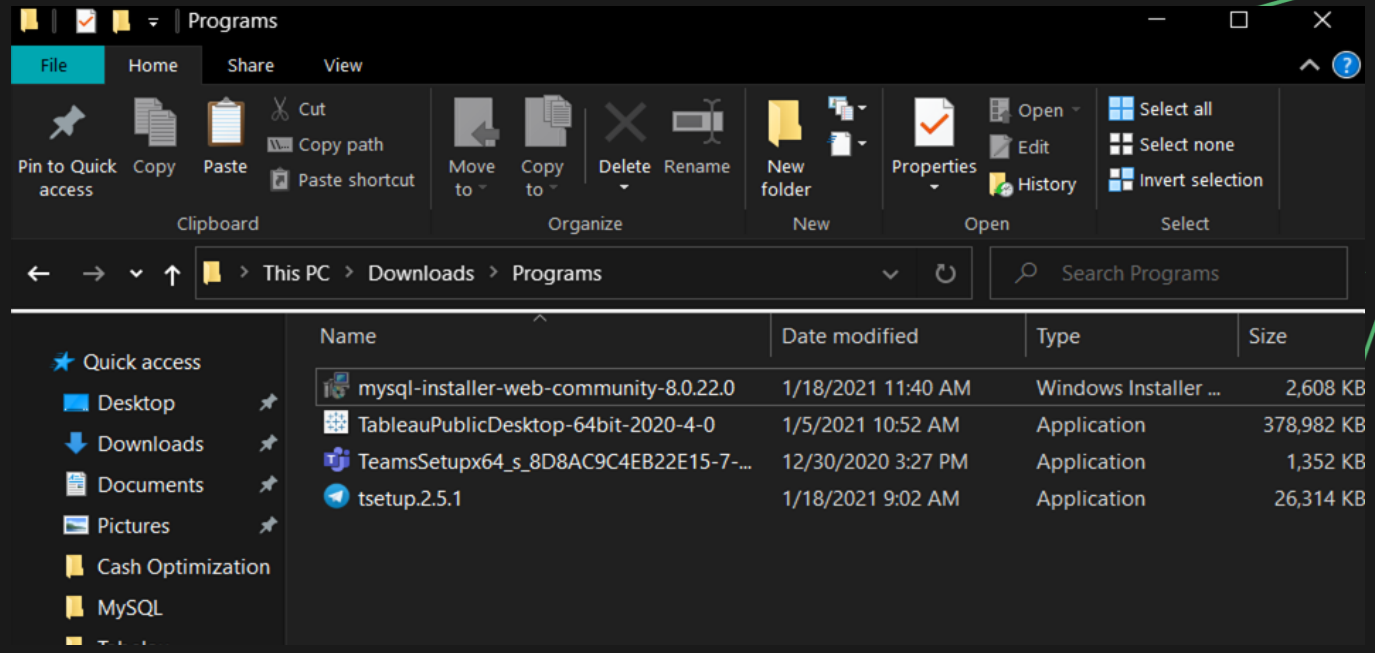




- Setelah anda klik download, MySQL akan menampilkan form login atau membuat akun di mysql.com. Jika berminat, silahkan isi form registrasi, namun jika tidak, lanjutkan dengan klik link “» No thanks, just start my download!”.
- Proses download akan berlangsung beberapa saat.

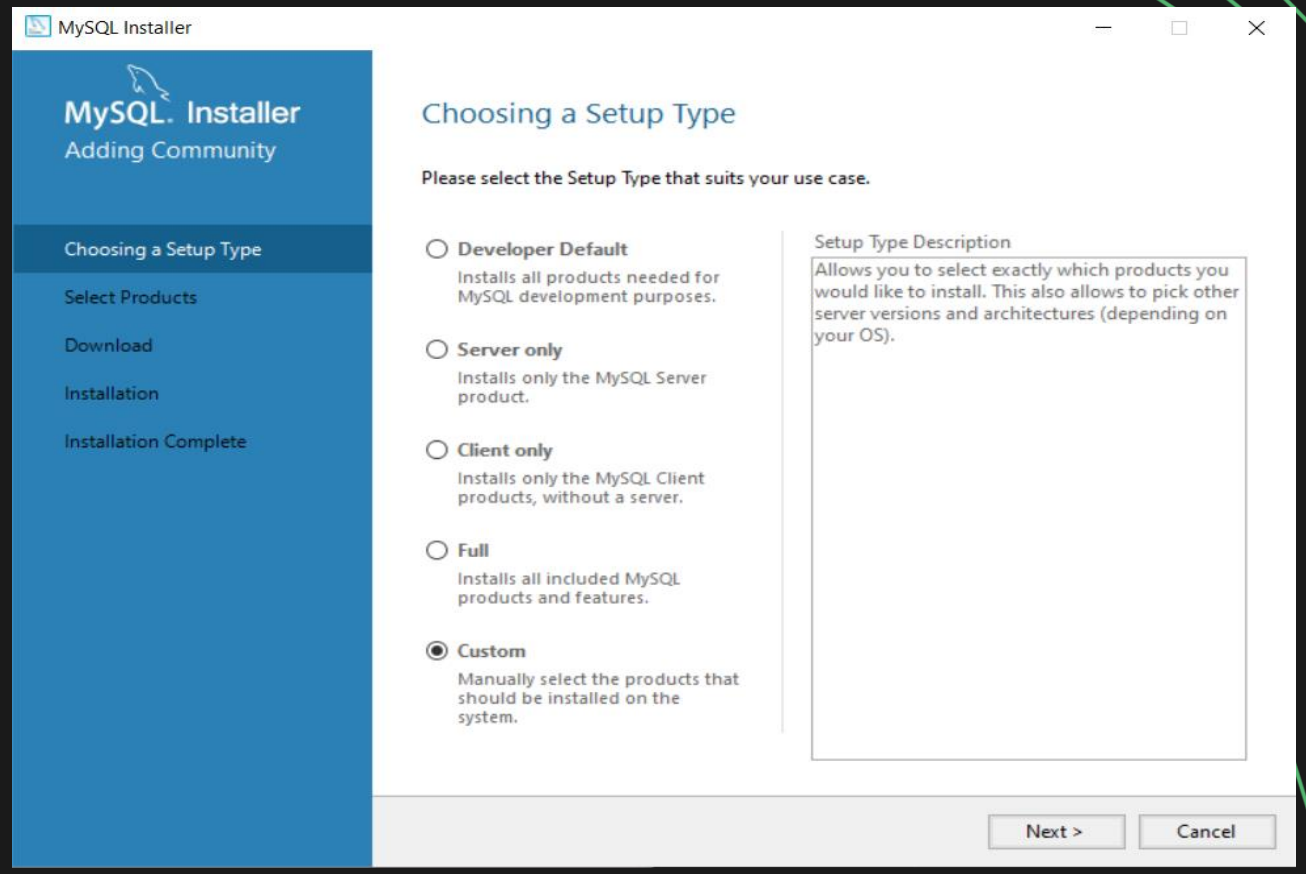
Installation Process

- Setelah proses download file MySQL selesai, kita akan mendapati file instalasi MySQL berupa: **mysql-installer-community-8.0.22.0.msi**.
- Kemungkinan versi yang anda download akan berbeda karena MySQL terus mengeluarkan versi terbaru, namun langkah-langkah instalasinya kurang lebih tetap sama.



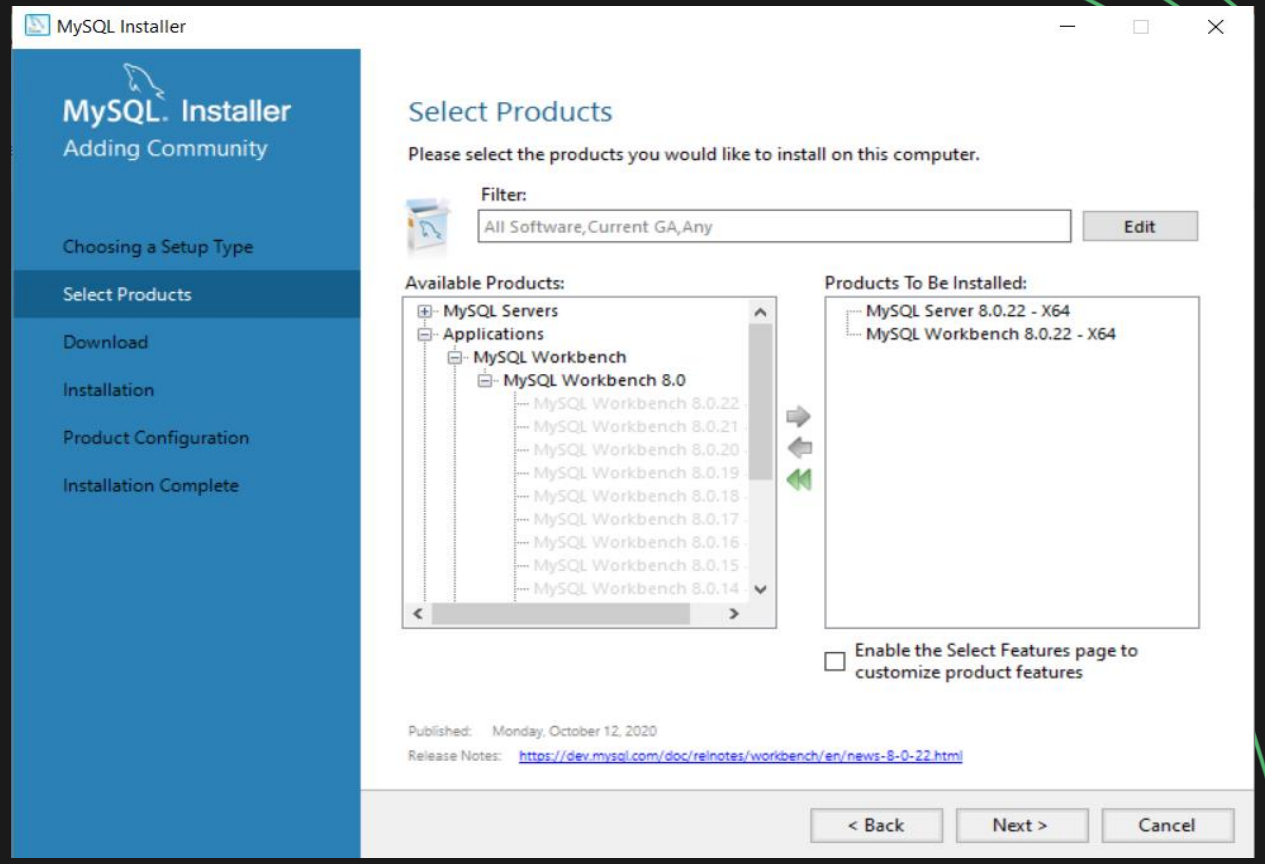
Installation Method

- Pada jendela “**Choosing a Setup Type**”, terdapat beberapa pilihan tipe instalasi. Mulai dari untuk developer default, server only, client only, full hingga custom.
- Kita akan memilih opsi "Custom" untuk kemudian memilih tools apa saja yang akan diinstall (MySQL server dan MySQL Workbench)



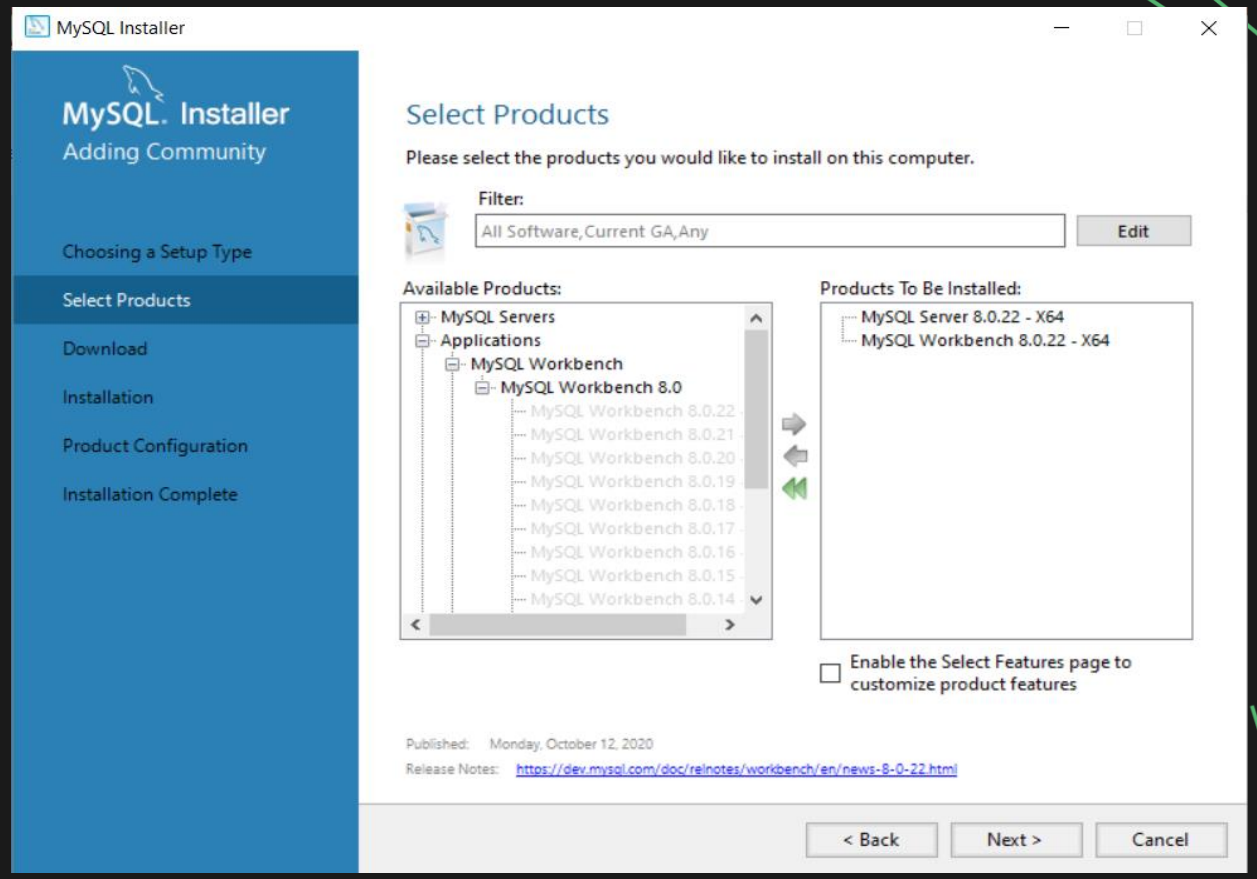
Add MySQL server

- Pada jendela “**Select Products and Features**” kita bisa memilih aplikasi dan fitur apa saja yang ingin diinstall. Kita akan install workbench dan sql server
- Silahkan klik tombol tanda tambah (+) paling atas (MySQL Server), sampai anda menemukan: **MySQL Server 8.0.22 – X64**. Pilih pilihan ini, lalu klik tombol panah hijau ditengah-tengah jendela untuk memindahkan dari jendela “Available Product” ke “Product/Features To Be Installed“.
- Jangan dulu klik "**next**" karena kita akan menambahkan MySQL Workbench



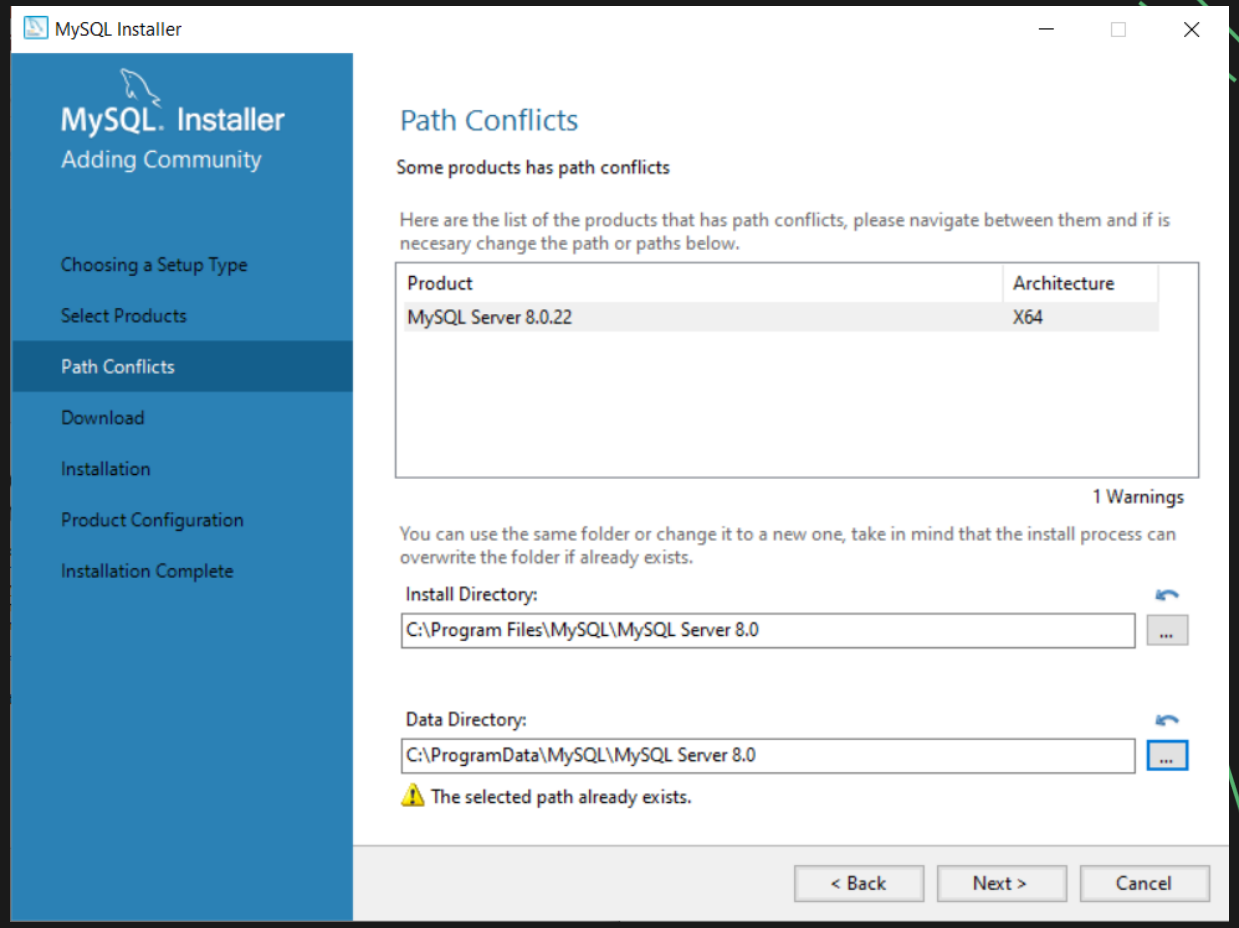
Add MySQL Workbench

- klik tombol tanda tambah (+) paling atas (MySQL Server), sampai anda menemukan: **MySQL Workbench 8.0.22 – X64**. Pilih pilihan ini, lalu klik tombol panah hijau ditengah-tengah jendela untuk memindahkan dari jendela “Available Product” ke “Product/Features To Be Installed“.
- Klik "**next**" untuk berpindah ke tahap selanjutnya



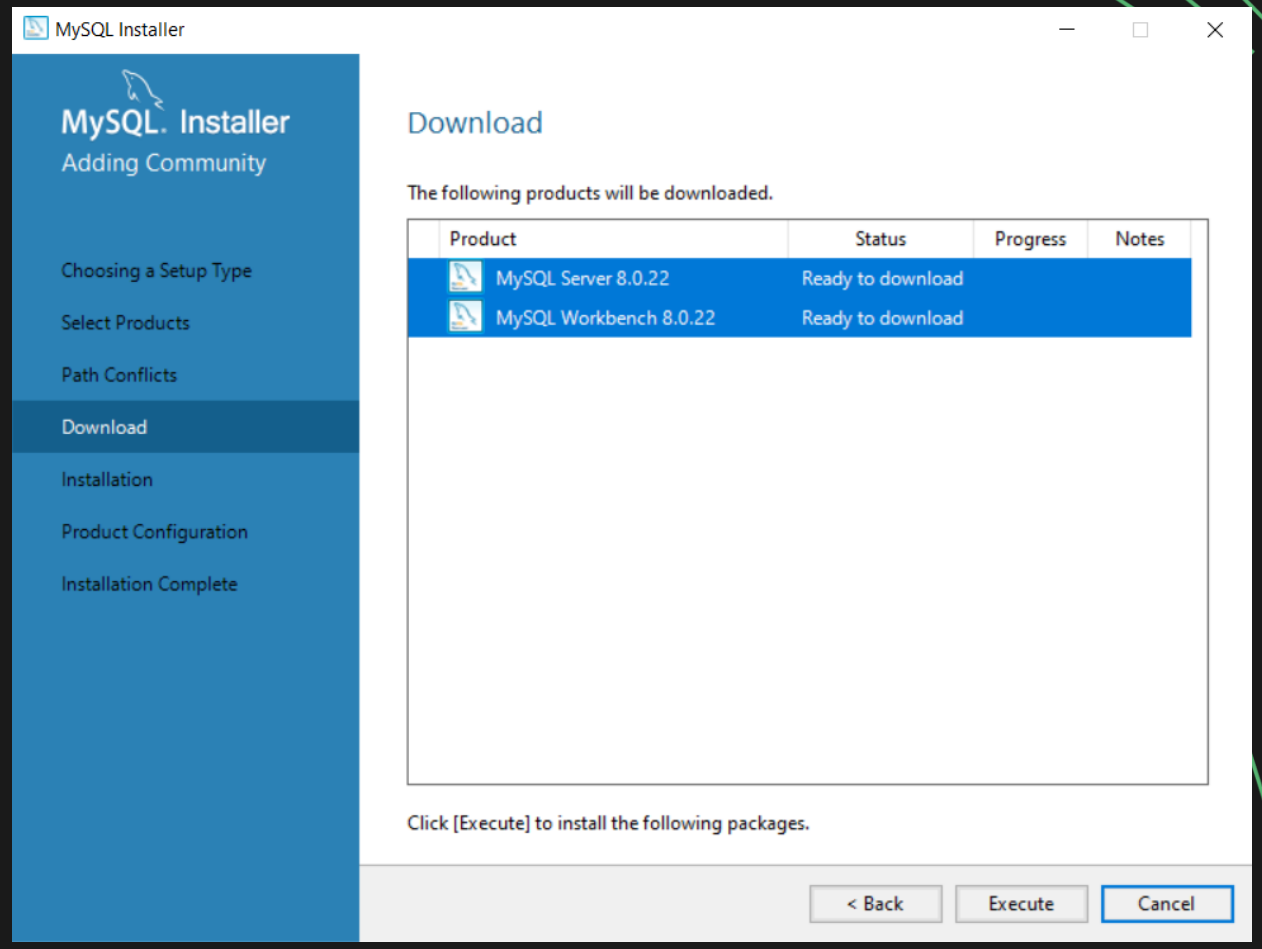
Add MySQL Workbench

- Pilih folder di mana anda akan menyimpan hasil installasi atau biarkan saja mengikuti setting default.
- Klik "**next**" untuk berpindah ke tahap selanjutnya



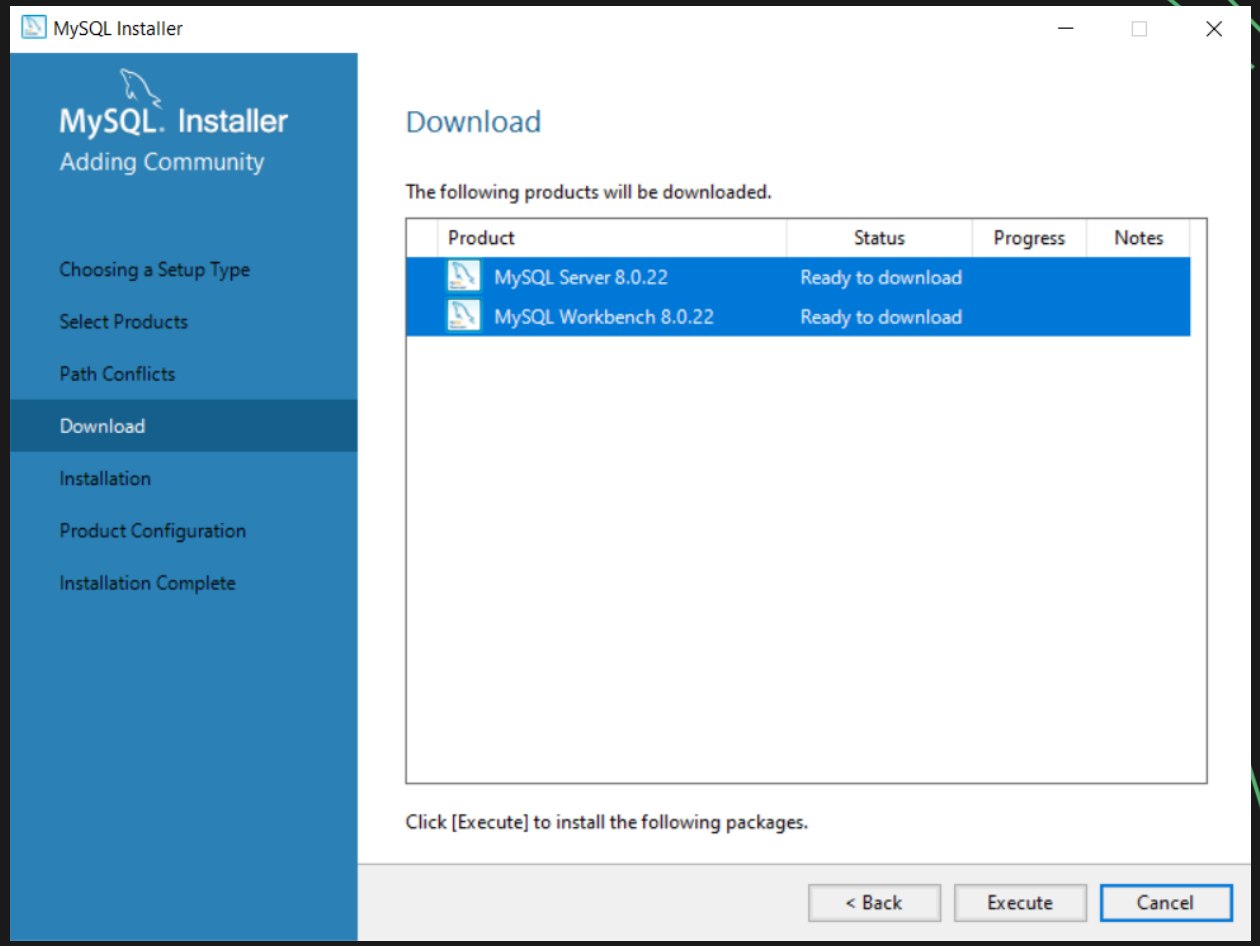
Download

- Pada Jendela **Download** ini, akan ditampilkan produk dan fitur apa saja yang sudah kita pilih sebelumnya.
- Kita akan download dua produk **MySQL Server 8.0.22** dan **MySQL Workbench 8.0.22**
- Silahkan klik tombol **Execute** untuk memulai proses download
- Proses download akan memakan waktu beberapa saat karena fitur yang akan kita install belum didownload sebelumnya
- ketika selesai, klik tombol Next untuk eksekusi instalasi



Execute Installation

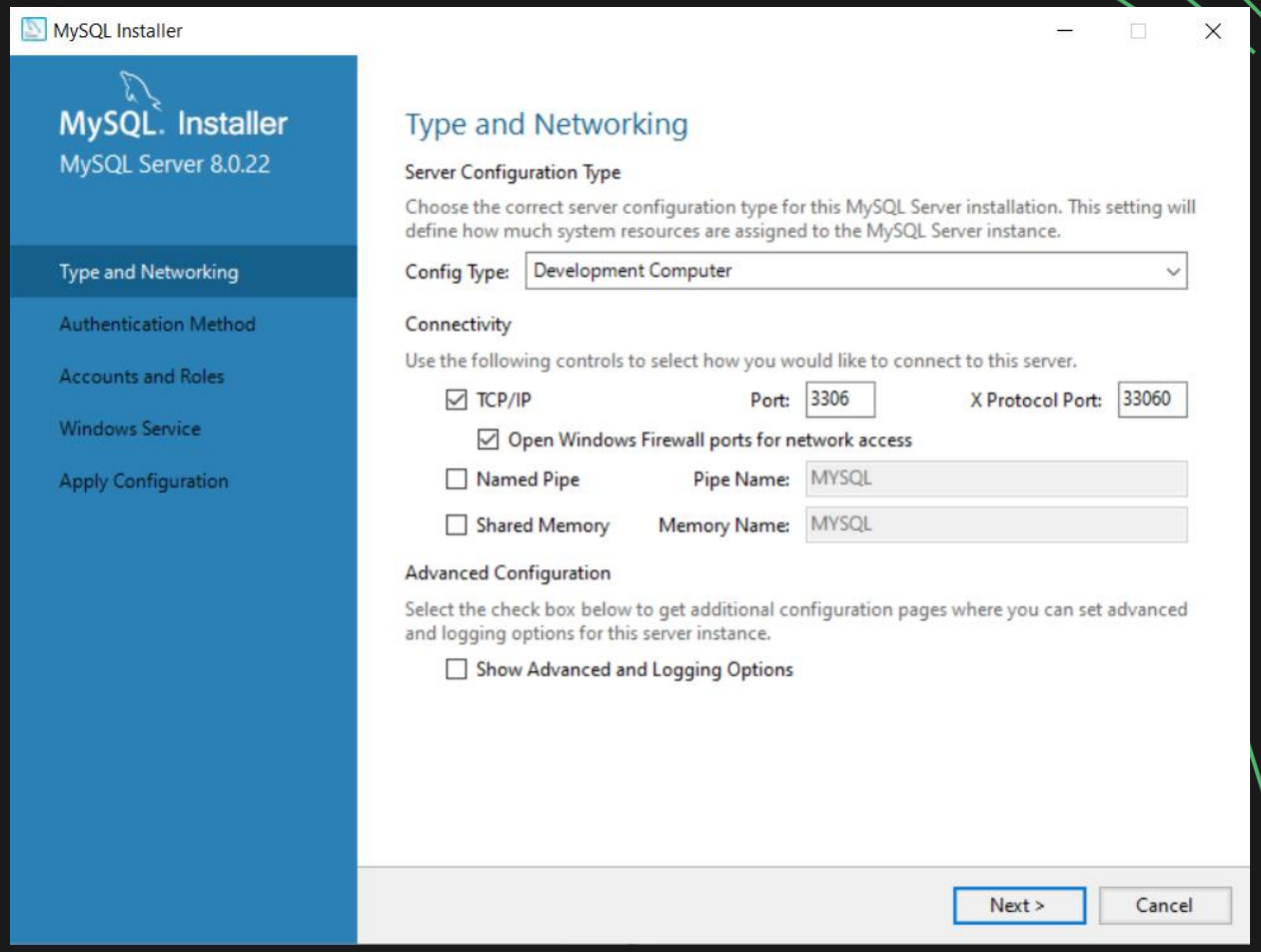
- Klik "Execute" untuk memulai instalasi dan tunggu hingga selesai
- ketika selesai, klik tombol Next beberapa kali hingga kita masuk ke bagian **konfigurasi awal MySQL**.



MySQL Configuration

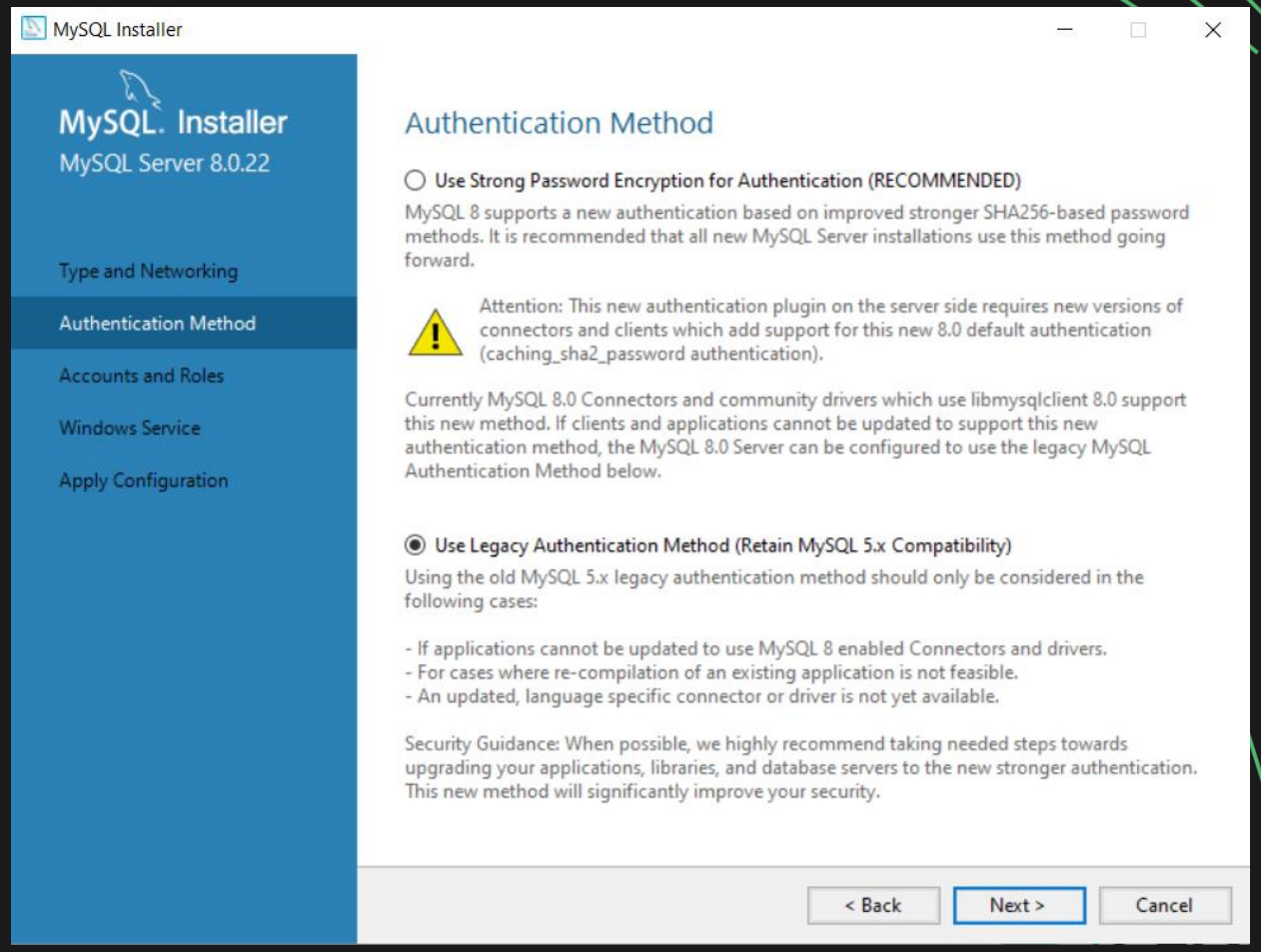
Type and Networking

- Biarkan pengaturan ini mengikuti pengaturan default
- klik “**next**”



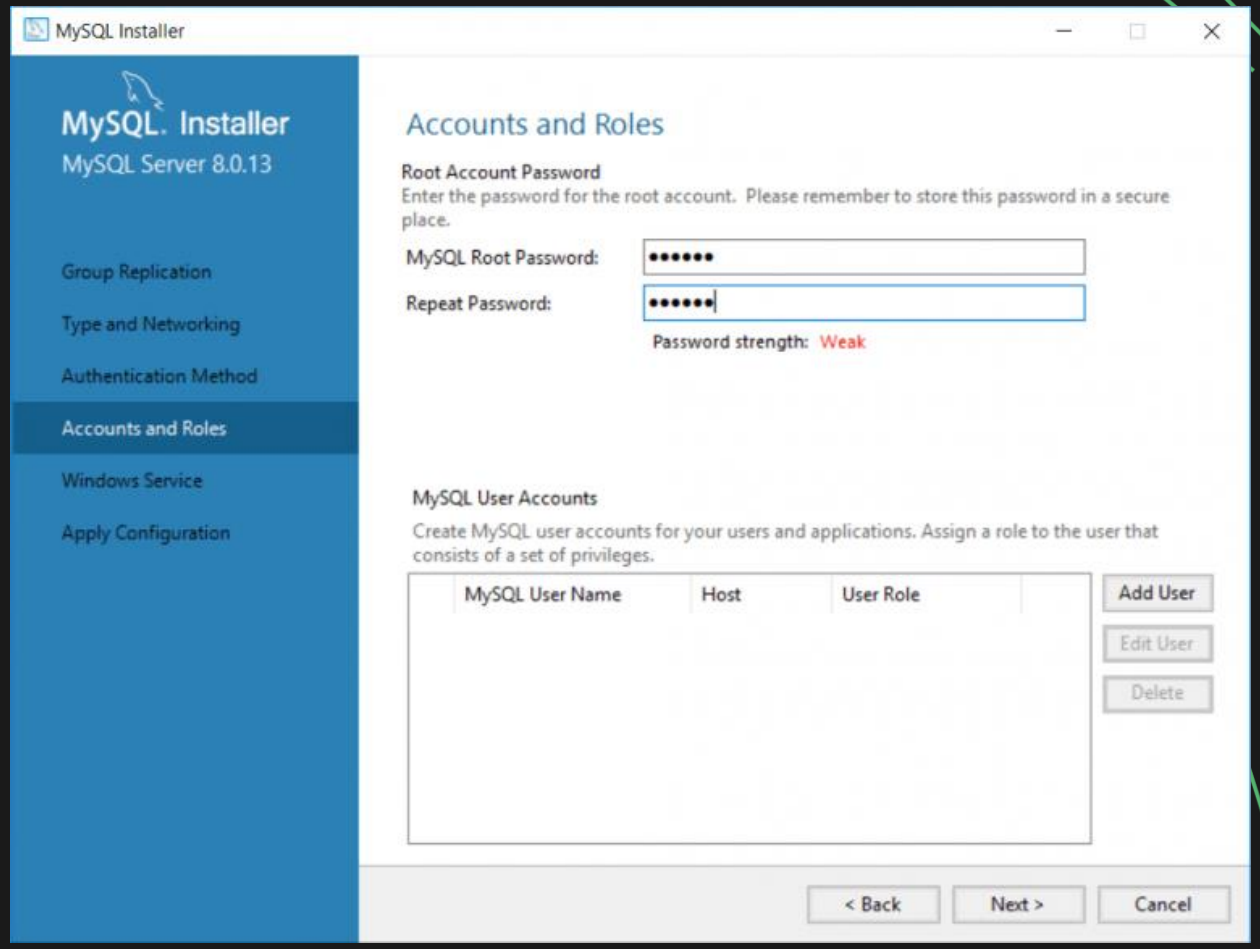
Authentication

- Pilih opsi “**Use Legacy Authentication Method**”
- klik “**next**”



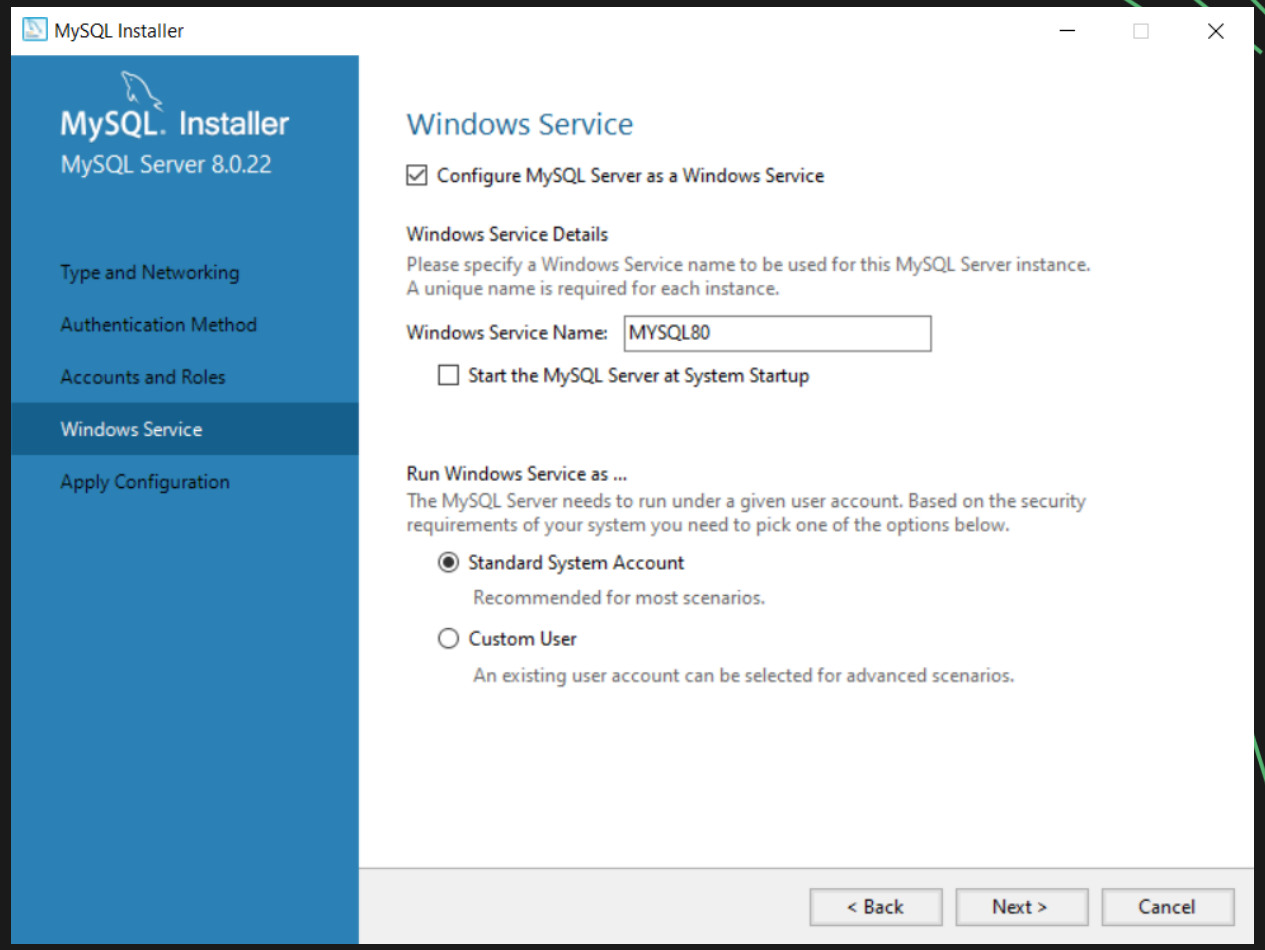
Password

- Jendela berikutnya adalah **Account and User Role**. Pada bagian ini kita akan membuat ***password*** untuk user **root** dan jika diinginkan juga bisa membuat user lain.
- Pada bagian atas, kita harus menginput *password* untuk user root. **Root** adalah user tertinggi di dalam MySQL, atau sering disebut juga dengan ***superuser***. User root memiliki hak akses kepada semua tabel dan database, termasuk menghapus seluruh database yang ada.
- Akhiri jendela ini dengan klik tombol "**Next**".



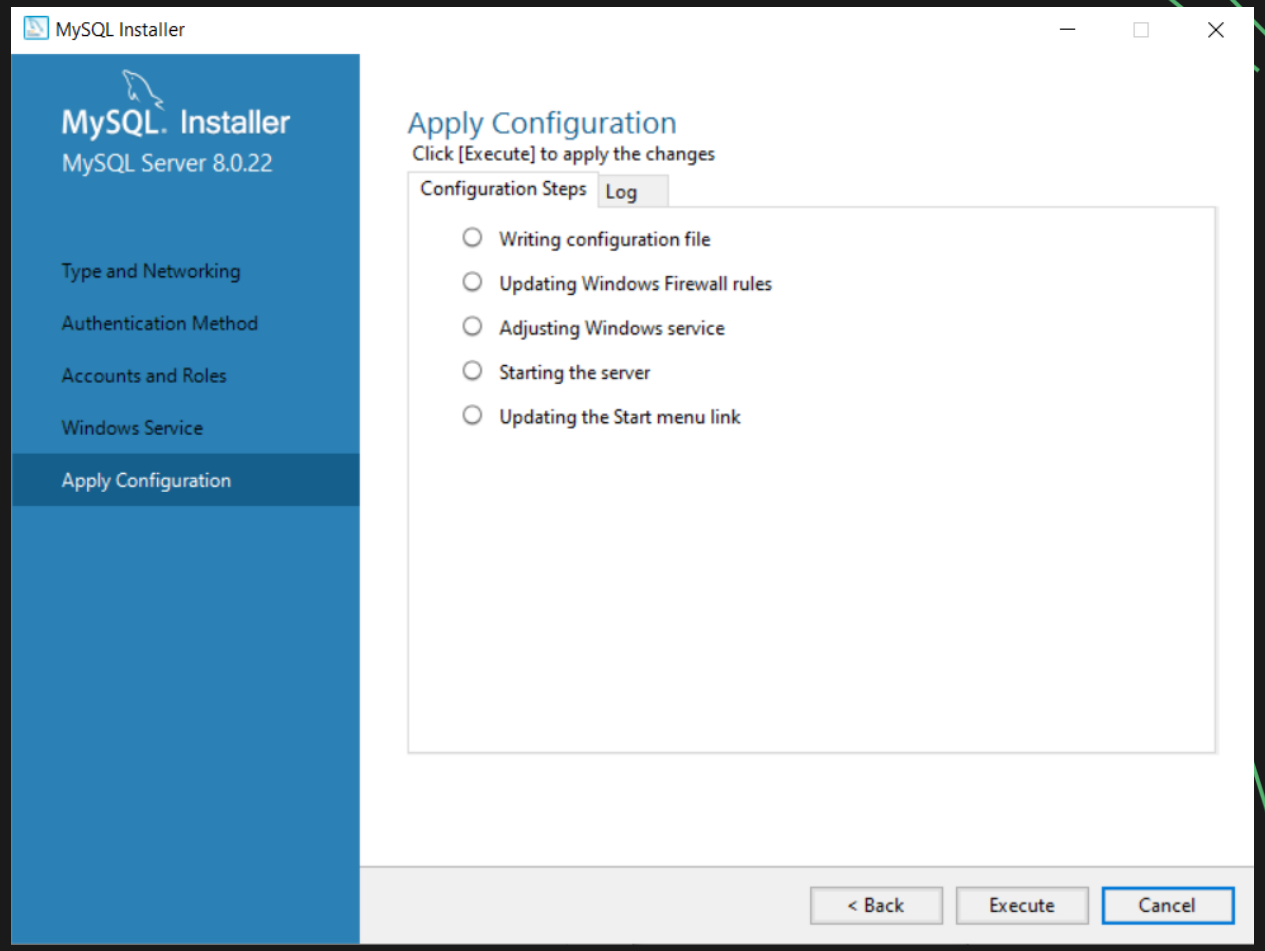
Windows Service

- Jendela berikutnya **sangat penting**, terutama jika anda berencana menginstall lebih dari 1 MySQL di dalam komputer (misalnya satu lagi dari XAMPP).
- Jika checkbox "*Configure MySQL Server as Windows Service*" dipilih, MySQL akan diinstall sebagai "**Windows Service**". Artinya, MySQL akan langsung aktif setiap Windows berjalan.
- Ini bisa menjadi masalah jika anda ingin menjalankan MySQL dari sumber lain (seperti XAMPP). Karena MySQL tidak bisa berjalan bersamaan di satu komputer (jika menggunakan port yang sama). Oleh karena itu hapus pilihan "*Configure MySQL Server as Windows Service*", lalu klik **Next**.

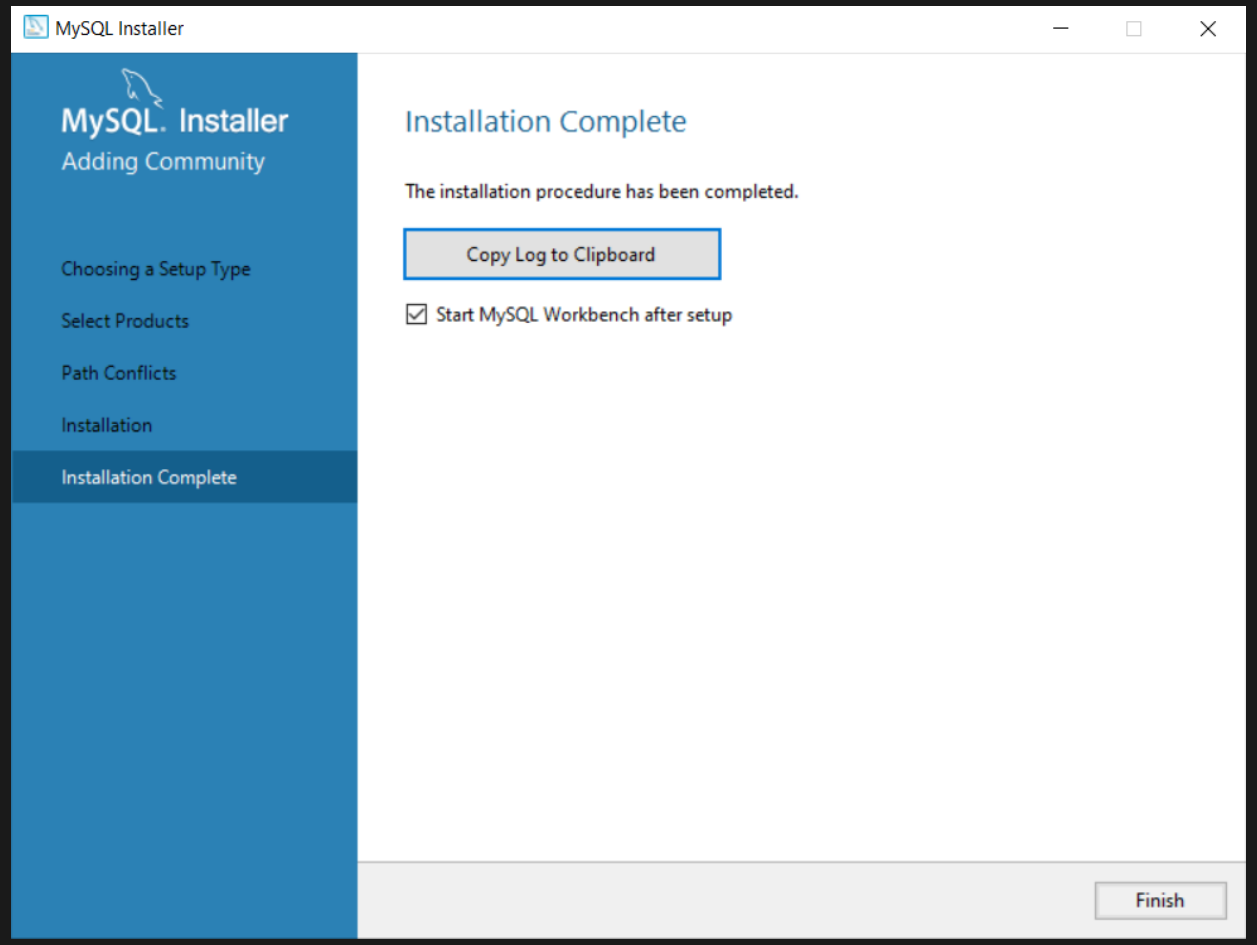


Apply Configuration

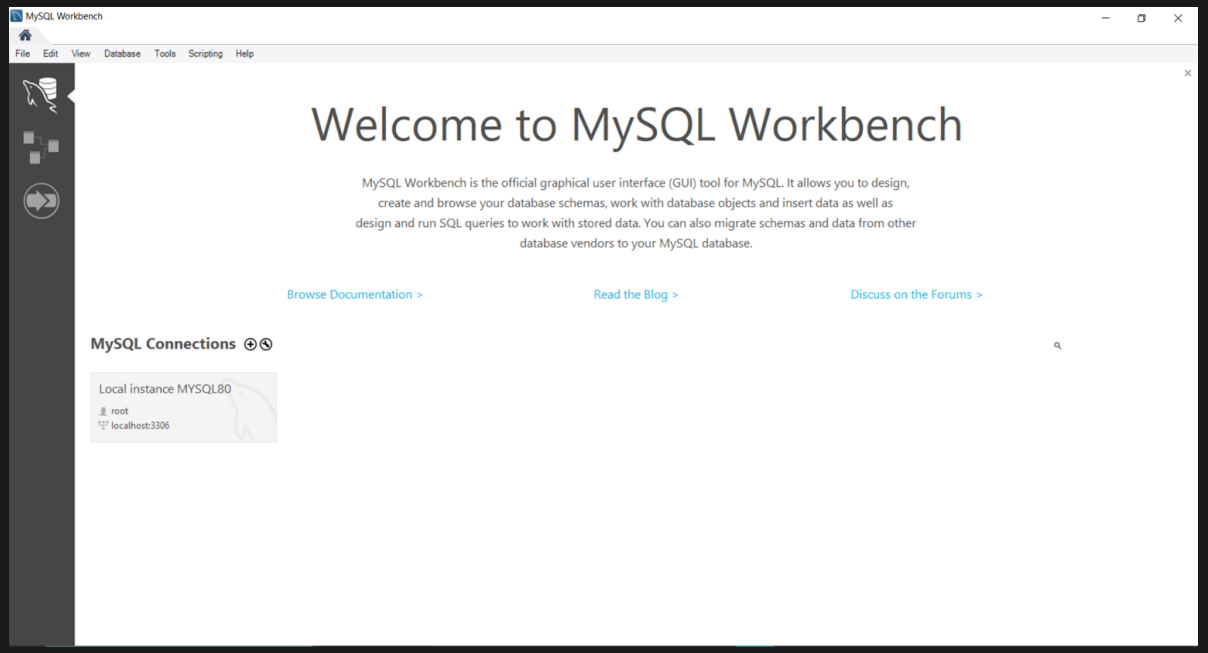
- Jendela terakhir adalah “**Apply Configuration**”.
- Di sini seluruh setingan yang sudah kita pilih akan diterapkan.
- Klik tombol **Execute**. Lalu klik “next” ketika selesai



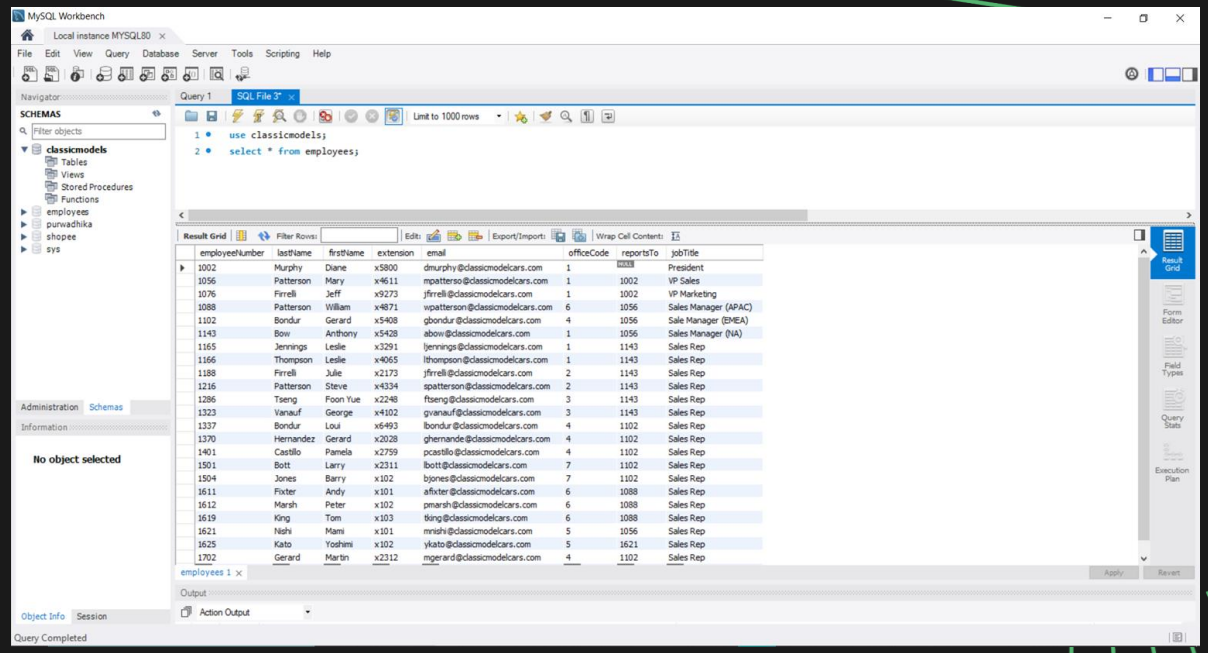
Installation Complete



Login to MySQL Workbench



Click a connection and insert the password



MySQL Workbench Interface

Thank You!

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